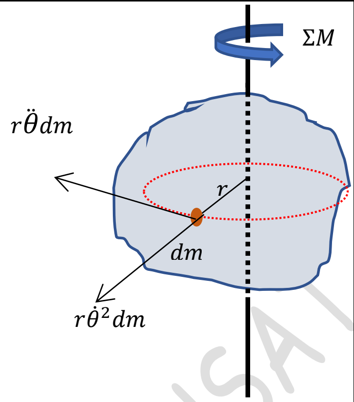
7.2 Kinetics of rotation

Equation of motion

Consider a rigid body rotating about a fixed axis under the action of a resultant moment ΣM. Let dm represents the mass of a particle at a radius r from the axis of rotation. This particle is moving along a circle. If the angular velocity and angular acceleration of the body are $\dot{\theta}$ and $\ddot{\theta}$, then the mass dm will experience a normal acceleration $r\dot{\theta}^2$ directed towards the centre of the circle and a tangential acceleration $r\ddot{\theta}$ tangential to the circle. Then by D'Alembert's principle, normal and tangential inertia forces of magnitude $r\dot{\theta}^2\, dm$ and $r\ddot{\theta}\, dm$ will be acting on the mass element in the opposite directions of accelerations. As the normal inertia force is intersecting with the axis, it won't produce any moment about the axis of rotation.



Moment of the tangential inertia force about the axis od rotation,
$= r\ddot{\theta}\, dm\ \times r = \ddot{\theta}\, r^2dm$

If we consider infinite number of such elemental masses, the total moment of all the tangential inertia forces = $\int \ddot{\theta}\, r^2dm$. By D'Alembert's principle, the system is in dynamic equilibrium. Hence, the algebraic sum of the external moments and inertia moments can be equated to zero.

$$\int \ddot{\theta}\, r^2dm - \Sigma M = 0$$

Since $\ddot{\theta}$ is the same for all the elemental masses, that can be taken outside the integral.

$$\ddot{\theta} \int r^2dm = \Sigma M$$

By definition, $\int r^2dm$ is the mass moment of inertia (I) of the body about the axis ofrotation.

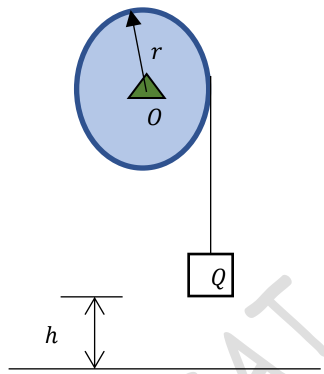
Therefore, the equation of rotational motion of a body about an axis can be written as

$$I\ddot{\theta}\ = \Sigma M$$

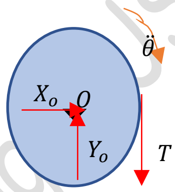
The external moment ΣM produce an angular acceleration $\ddot{\theta}$ in its direction.

Example 1

A solid right circular drum of radius $r = 30\ cm$ and weight $W = 145\ N$ is free to rotate about its geometric axis. A flexible cord carrying a weight of $Q = 45\ N$ is wound around the circumference of the drum. If the weight Q is released from rest, find the time taken by the weight to fall through a distance of $h = 3m$. With what velocity v will it strike the floor?



Let us consider the free-body diagram of the drum. It is rotating about its geometric axis in the clockwise direction with an angular acceleration $\ddot{\theta}$. The forces acting on the drum are the reaction components X_o and Y_o at the support and the tension T in the cord.



As the drum is rotating about its axis through O, we can write the equation of rotation as

$$I_o\ddot{\theta} = \Sigma M_o$$

Mass moment of inertia of the circular drum of weight W and radius r about its geometrical axis through O

$$I_o = \frac{W}{g}\frac{r^2}{2} = \frac{145}{9.81} \times \frac{0.3^2}{2} = 0.665\ kg.m^2$$

Moment of all forces about the axis of rotation in the direction of angular acceleration,

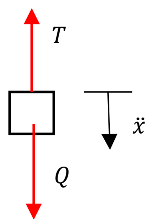
$$\Sigma M_o = T \times r = 0.3\ T\ Nm$$

$$0.665\ \ddot{\theta} = 0.3T$$

Now let us consider the free-body diagram of the block, which is undergoing a rectilinear translation. The forces are the weight Q and the tension T. It is moving down with an acceleration $\ddot{x}$.

The equation of motion of the block,

$$\frac{Q}{g}\ddot{x} = \Sigma F = Q - T$$



Now we have two equations in three unknowns. We can also establish a relation bwtween $\ddot{x}$ and $\ddot{\theta}$

When the drum turns through an angle θ, the length of the cord unwinding from the drum $x = r\theta$.

$$\therefore \ddot{x} = r\ \ddot{\theta}$$

$$\frac{Q}{g}\ddot{x} = Q - \frac{0.665\ \ddot{\theta}}{0.3} = Q - \frac{0.665}{0.3} \times \frac{\ddot{x}}{0.3}$$

$$\ddot{x} = 3.76\ m/s^2$$

The tension in the string, $T = Q\left(1 - \frac{\ddot{x}}{g}\right) = 27.75N$

Work-energy principle applied to the block.

Initial kinetic energy of the block=0

Let v be the velocity when it strikes the floor. Its final kinetic energy$=\frac{Q}{g}\frac{v^2}{2}$

Change in kinetic energy $= \frac{Q}{g}\frac{v^2}{2} - 0$

Work done by the forces in travelling h distance , $= (Q - T) \times 3m = 51.74\ Nm$, Therefore, by work energy principle, change in kinetic energy= net work done

$$\frac{Q}{g}\frac{v^2}{2} - 0 = 51.74$$
$$v = 4.75\ m/s$$

Impulse-momentum principle

Initial momentum of the block =0

Final momentum of the block=$\frac{Q}{g}\ v = 21.78\ kg\ m/s$

Change in momentum of the block= $21.78 - 0 = 21.78\ kgm/s$

Impulse of the forces = $\int_0^t (Q - T)dt = 17.25\ t$

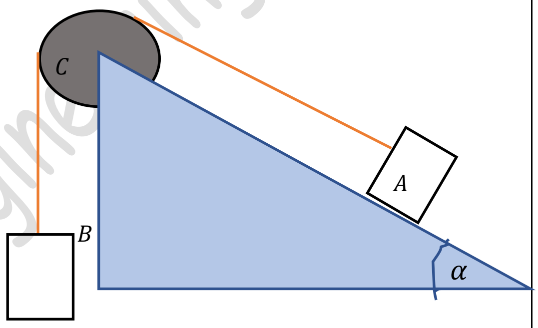
Hence, by impulse-momwntum priciple, change in momentum= impulse of the forces

$$21.78\frac{kgm}{s} = 17.25t$$
$$t = 1.26\ s$$

You can also find out the values of v and t by considering the kinematics of the block Q.

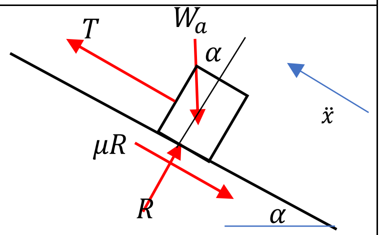
Example 2

The figure below represents a system consisting of a block A of weight $W_a = 3900\ N$ connected to a block B of weight $W_b = 2900\ N$ by a flexible cord which over runs a pulley of radius $r = 0.3$ and weight $W_c = 1450\ N$. The block B moves down and block A slides along a plane inclined to the horizontal by the angle $\alpha = 30^0$ and for which the coefficient of friction $\mu = 0.1$. Friction on the axle of the pulley is assumed to supply a constant resisting torque of $13.7\ Nm$. Assuming that no slipping occurs between the cord and the pulley, determine (a) the acceleration of the blocks and (b) the maximum tensile force in the cord



Let the block B moves down with acceleration $\ddot{x}$. Since the string is inextensible, the acceleration of block A is also $\ddot{x}$. The friction in the pulley is not negligible, the tension in the string is not the same.

The pulley rotates with angular acceleration $\ddot{\theta}$. As there is no slipping, we can write $x = r\theta \quad \ggg \quad \ddot{x} = r\ddot{\theta}$



From the free-body diagram of the block A,

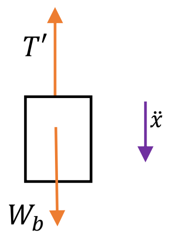
$$R - W_a \cos\alpha = 0 \qquad \ggg \qquad R = 3900 \cos 30 \ggg \quad R = 3378\ N$$

The equation of motion along the plane is

$$\frac{W_a}{g}\ddot{x} = T - \mu R - W_a \sin\alpha \ \ggg\ \ T = \frac{3900}{9.81}\ddot{x} + 0.1 \times 3378 + 3900 \sin 30 \ggg \quad T = 397.6\ \ddot{x} + 2287.8$$

Now consider the motion of block B.

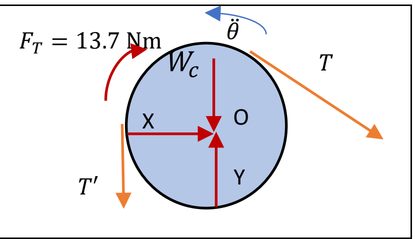
$$\frac{W_b}{g}\ddot{x} = W_b - T' \quad \ggg\gg \quad T' = 2900 - 295.6\ \ddot{x}$$



Consider the rotation of the pulley.

The equation of motion is $I_o\ddot{\theta} = \Sigma M_o$

$$\frac{W_c}{g}\frac{r^2}{2}\ddot{\theta} = T'r - Tr - 13.7$$



$$\frac{1450}{9.81}\times\frac{0.3^2}{2}\times\frac{\ddot{x}}{0.3} = 0.3\,T' - 0.3\,T - 13.7$$

$$T' - T = 73.9\,\ddot{x} + 45.7$$

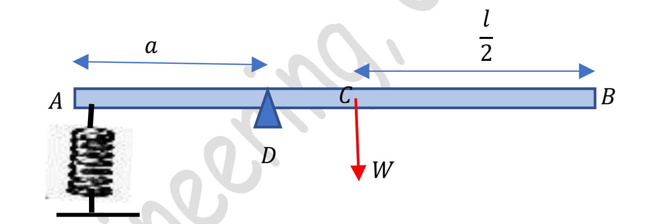
$$i.e.,\quad (2900 - 295.6\,\ddot{x}) - (397.6\,\ddot{x} + 2287.8) = 73.9\,\ddot{x} + 45.7$$

$$\ddot{x} = 0.74\,m/s^2$$

$$T = 2582\,N \quad and\ T' = 2681N$$

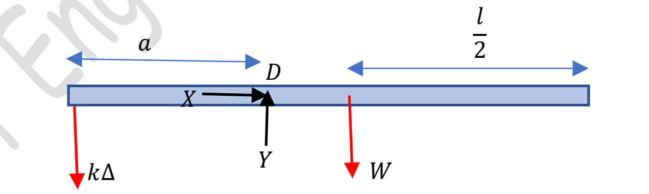
Example 3

A slender prismatic bar of weight W and length l is hinged at D and held in a horizontal position of equilibrium by a spring of constant k. Find the period of small rotational oscillations of the bar in the vertical plane of the figure. Given $W = 22.5\,N$, $l = 0.9\,m$, $a = 0.3m$, $k = 900\,N/m$

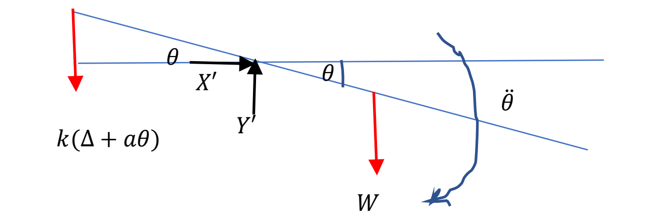


Consider the bar in the static equilibrium position. The spring will be having an initial deflection Δ.

$$\Sigma M_D = 0 \ggg\ k\Delta a - W\left(\frac{l}{2} - a\right) = 0$$



Now, let us assume that the bar is disturbed from the equilibrium position. It will make small rotational oscillations. Consider an instant when the bar makes an angle θ as shown. Let the angular acceleration of the bar be $\ddot{\theta}$ in the clockwise direction at this instant of time.



The end A of the bar will have a displacement $x = a\theta$. So the spring will have a net elongation of $\Delta + a\theta$. Let us write the equation of motion of the bar.

$I_D\,\ddot{\theta} = \Sigma M_D$ (the moment should be in the direction of angular acceleration.)

Mass moment of inertia of the slender bar about its centroidal axis, $I_C = \frac{W}{g}\frac{l^2}{12}$.

By applying the parallel axis theorem, $I_D = \frac{W}{g}\frac{l^2}{12} + \frac{W}{g}\left(\frac{l}{2} - a\right)^2 = 0.206\,kg.m^2$

$$0.206\,\ddot{\theta} = W\left(\frac{l}{2} - a\right) - k(\Delta + a\theta)a = W\left(\frac{l}{2} - a\right) - k\Delta a - ka^2\theta$$

$$I_D\ddot{\theta} = -ka^2\theta$$

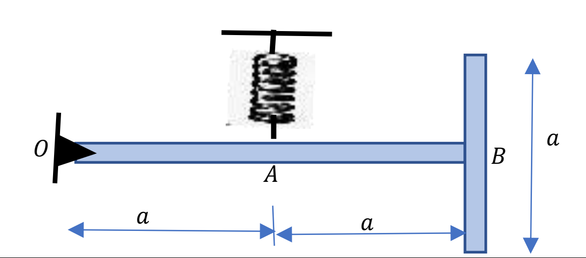
$$\ddot{\theta} = -\left(\frac{ka^2}{I_D}\right)\theta = -393.3\,\theta \qquad \ggg\gg\ \ddot{\theta} + 393.3\,\theta = 0$$

This equation is now in the form $\ddot{\theta} + p^2\theta = 0$, where $p = \sqrt{393.3} = 19.83\,s^{-1}$

The period of rotational oscillations, $T = \frac{2\pi}{p} = 0.317\,s$

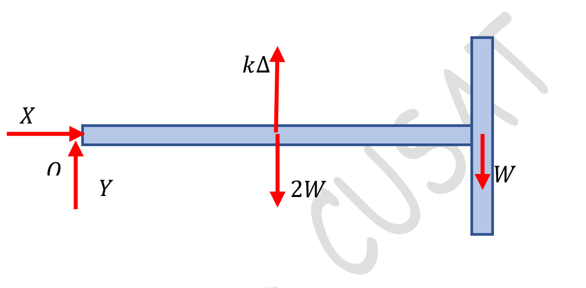
Example 4

A slender T-bar of uniform cross section and total weight $3W$ is supported in a vertical plane by a hinge at O and a spring of constant k at A. Find the period of small oscillations in the plane of the figure.



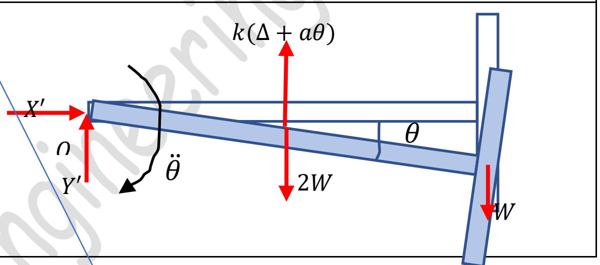
Since, the T-bar is made of two uniform bars in which one bar is having twice the length of the other, the weight of the longer bar will be $2W$ and the shorter bar is W.

Let us first consider the free-body diagram of the T-bar in the static equilibrium position. The spring will be having a static deflection Δ



$$\Sigma M_o = 0 \ \ggg\gg \ k\Delta \times a - 2W \times a - W \times 2a = 0$$

Now, let us consider that the T-bar is disturbed from its equilibrium position. It will make small rotational oscillations about the axis of rotation passing through O. Consider an instant when the bar is rotating clockwise through an angle θ with an angular acceleration $\ddot{\theta}$



The spring will be now having a deflection of $\Delta + a\theta$. Writing down the equation of rotation of the T-bar,

$$I_o\ddot{\theta} = \Sigma M_o$$

The mass moment of inertia of the T-bar about the perpendicular axis through O, $I_o = I_{1o} + I_{2o}$

$$I_{1o} = \frac{(2W)}{g}\frac{(2a)^2}{12} + \frac{(2W)}{g}a^2 = \frac{8Wa^2}{3g} \quad \lll\ggg \quad I_{2o} = \frac{W}{g}\frac{a^2}{12} + \frac{W}{g}(2a)^2 = \frac{49Wa^2}{12g}$$

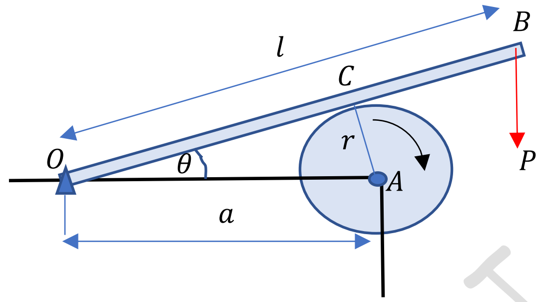
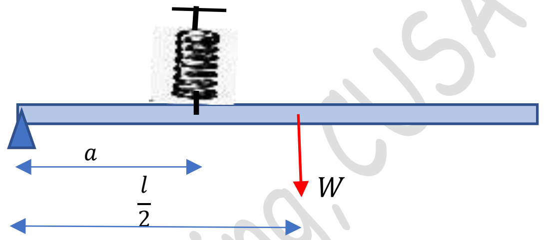
$$I_o = I_{1o} + I_{2o} = \frac{8Wa^2}{3g} + \frac{49Wa^2}{12g} = \frac{27Wa^2}{4g}$$

$$I_o\ddot{\theta} = \Sigma M_o \quad \gg\gg \quad \frac{27Wa^2}{4g}\ddot{\theta} = -k(\Delta + a\theta)a + 2Wa + W2a = (-k\Delta a + 2W \times a + W \times 2a) - ka^2\theta$$

$$\ddot{\theta} = -\frac{4kg}{27W}\theta \quad \ggg\gg \quad \ddot{\theta} + \frac{4kg}{27W}\theta = 0$$

The equation is of the form $\ddot{\theta} + p^2\theta = 0$ where $p = \sqrt{\frac{4kg}{27W}}$

Therefore, the period of small rotational oscillations, $T = \frac{2\pi}{p} = 2\pi\sqrt{\frac{27W}{4kg}} = 16.32\sqrt{\frac{W}{kg}}$

#Exercise 1 A right circular drum of radius $r = 40\ cm$ and weight $W = 1800\ N$ rotating at $600\ rpm$ is braked by the device as shown. Find the time required to bring the drum to rest if the coefficient of friction between the drum and the braking bar is 0.25 if $l = 120\ cm, a = 75\ cm$ and $P = 450\ N$.	
#Exercise 2 A slender prismatic bar AB of weight $W = 25\ N$ and length $l = 0.9\ m$ is hinged at A and supported in a horizontal position by a spring at D. Find the period of small rotational oscillations of the bar in the vertical plane, if $k = 900\ N/m$ and $a = 0.3\ m$	
#Exercise 3 A solid right circular drum of weight $W = 720\ N$ and radius $r = 0.3m$ can rotate without friction about its horizontal geometric axis. A flexible but inextensible belt overhanging the drum carries a weight $Q = 180\ N$ at one end and is attached to a spring of constant $k = 10800\ N/m$ at the other end. Assuming that there is no slip between the belt and the drum, find the period of small oscillations of the system.